ActInf Livestream #051.0 ~ "Canonical neural networks perform active infe

Livestream | *1:22:11*

All right.

Welcome. It is ActInf livestream number 51.0. This is the background and context first discussion for Canonical Neural Networks Perform Active Inference by Isomura et al. It's October 26, 2022.

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Today, it's the first of several discussions that we'll have on the paper, Canonical Neural Networks Perform Active Inference, 2022, by Takuya Isomura, Hideaki Shimizaki, and Carl Fristen.

The video is just an introduction to some of the ideas. It's not a review or final word. There will be an overview of the structure of the paper, and then we'll go through many of the key points.

Also, just to disclaim, there's many, many other and better resources to learn about neural networks. So I would very much welcome those with a technical understanding of neural networks and or some of the more applied computational or otherwise aspects of neural networks.

It would be awesome to have them on for the dot one and dot two because it was not an area I was familiar with.

And so hope that I can hear more from the authors in our coming weeks and others.

I'm Daniel. I'm a researcher in California, and this will just be a solo dot zero, which I guess hasn't happened in a while.

So, in the making of this, here's some of the generated art prompts. Area 51 active inference, Area 51 neural network, Area 51 active inference, and Area 51 active inference neural network.

Just some interesting images coming out of stable diffusion.

So as to some big questions that the paper is addressing and that one might be interested in to come to the paper.

How can artificial neural networks be understood as generic optimization processes?

And what is the correspondence between neural dynamics and modern statistical inference methods?

Other big questions are about the history and next steps of the enmeshment of natural intelligence,

e.g. neuroscience and artificial intelligence, as well as, of course, whether to even play into this kind of distinction at all and have different integrated intelligence frameworks.

And one paper where any of the authors or anyone who has kind of resonated with this work is recent by Zador et al, 2022,

Towards Next Generation Artificial Intelligence Catalyzing the Neuro-AI Revolution.

And so this is a bunch of authors. And so it's interesting just to quote in terms of what some areas of discourse are saying right now,

which is neuroscience has long been an important driver of progress in artificial intelligence, AI.

We propose that to accelerate progress in AI, we must invest in fundamental research in neuro AI.

So that's one way to lead some of the developments that are happening in the paper we'll discuss.

What does it mean to be particular but generic?

That's a phrase used in the paper. So maybe that's kind of a jumping off point.

And then how can active inference help us understand the past, present and future?

Here of the interface with neural networks, statistics and neuro AI.

Here's the abstract.

This work considers a class of canonical neural networks comprising rate coding models,

where neural activity and plasticity minimize a common cost function.

And plasticity is modulated with a certain delay.

We show that such neural networks implicitly perform active inference and learning to minimize the risk associated with future outcomes.

Mathematical analyses demonstrate that this biological optimization can be cast as maximization of model evidence,

or equivalently minimization of variational free energy, under the well-known form of a partially observed Markov decision process model.

This equivalence indicates that the delayed modulation of Hebbian plasticity accompanied with adaptation of firing thresholds is a sufficient neuronal substrate to attain Bayes optimal inference and control.

We corroborated this proposition using numerical analysis of Bayes tasks.

This theory offers a universal characterization of canonical neural networks in terms of Bayesian belief updating and provides insight into the neuronal mechanisms underlying planning and adaptive behavioral control.

Here is the roadmap.

I'll be driving on the right side of the road.

Here is the known

Bayesian methods and start with a more theoretical and mathematical background and then eventually

present some maze simulations that are in MATLAB. Then there's a discussion and then the methods section is following the discussion and it has the subsections generative model, variational free energy, inference and learning, neural networks, simulations, and data analysis. And I think that I have kept to the right color codes consistently from here on forward. The black quotes are the direct quotes from the paper. Blue quotes are related to perception, orange for action, complete class theorems and neural networks together in purple. And then red is commentary. And then the highlighting for red is related to the variational methods, but it's kind of red-ish.

Okay. The paper's canonical neural networks perform active inference by the authors listed previously. It's in communications biology from 2022. And the key aims of the paper are laid out well in the first paragraph. So that's in black and other colors. And here the red is just kind of getting into initial conversation with the paper and connecting a little bit more to some previous streams before we go more into the specifics of the paper. The sentient behavior of biological organisms is characterized by optimization. Biological organisms recognize the state of their environment by optimizing internal representations of the external environmental dynamics generating sensory inputs. So that's the sensory recognition model. In addition, they optimize their behavior for adaptation to the environment, thereby increasing their probability of survival and reproduction. So that is bringing the entire active and inactive control theory, formal theories of action, action selections, planning, and planning as inference. All of these action-oriented, pragmatic turn-type ideas come into play when this action orange aspect is brought in. And it is slash must be slash should be, etc. oriented towards survival and persistence. Otherwise, we don't see that kind of thing for long over appropriate definitions for thing and long, etc. And this few sentences that the paper begins with summarizes a common theme in act imp, which is that there's basically two pathways towards proficiency in the niche. There's changing the mind, which is the perceptual and learning, and then there's changing the world through action. And that kind of integration of inference, cognitive inference, whether it's perceptual or learning or other, and action as enacted is part of the act imp, formulism, and shared with other areas. This biological self-organization is typically formulated as the minimization of cost functions wherein a gradient descent on a cost function furnishes neural dynamics and synaptic plasticity. So they are working with a framework where the way that this perception action flow is tractable computationally, either just for our current computers so that we can implement simulations and data fitting, or whether more fundamentally like this is the computational context of action, they're suggesting there's a way to think of it in terms of a cost function minimization. And also note like minimization maximization in a way they're interchangeable because there's just sometimes negative signs that can flip them. So it's the same energy and fitness landscape that is being navigated, whether you're minimizing to the bottom of the bowl and thinking of action selection that way and inference, or whether you're climbing to the top of the hill, gradient descent, gradient ascent, they're both kind of two sides of the same coin.

Two fundamental issues remain to be established, namely, so this is what the paper is saying, they're filling the gap in literature, the characterization of the dynamics of an arbitrary

neural network as a generic optimization process, and the correspondence between such neural dynamics and statistical inference found in applied mathematics and machine learning. The present work addresses these
issues by demonstrating that a class of canonical neural networks of rate coding models is functioning as,
and thus universally characterized in terms of, variational Bayesian inference under a particular
but generic form of the generative model. This is maybe the only slide that has resources from
the neural networks. This is the only slide that has resources that are available to the top of the
neural networks. This is the only slide that has a little bit of a little bit of a little bit of a
neural network. This is the only slide that has a little bit of a little bit of a little bit of a
search on a very common search tool resulted in 5-plus million results for what are neural networks.
And the first one was 3Blue1Brown from 2017, which is just a nice YouTube video as they very often make with these
beautiful renditions. And it's a very great video and there's many many others. A neural network is a
network or circuit of biological neurons and that is variously purely computational and or biological
map territory stuff. And they can be thought of as nodes and edges which are the firing rates
of the nodes. And the connections between the biological neurons being modeled as weights between nodes.
A positive weight reflects an excitatory connection while negative values mean inhibitory connections.
And here is data coming in and ending up activating different neurons in this final layer.
So this is kind of doing inference in a neural net work and then there's an action part. Okay.
The paper says, Variational Bayesian Inference offers a unified explanation for inference, learning, prediction,
decision making, and the evolution of biological form, which can be considered over multiple timescales.
So this returns to the earlier theme of like perception, action, and persistence within and among generations.
And the citations that are provided here for this are Friston Kilner Harrison, 2006 and Friston 2010.
Both very classic ACDEMF slash FEP papers, just to show one figure from each. Here's from the first citation
that's a question with a winged snowflake where if the snowflake ends up being somewhere that's too warm,
that's not able for it to maintain the structure, then it melts, melts into a dew, etc.
And it must be acting as if in one way or another, it's staying above that phase boundary.
That's how it's statistically identified. It's how it's ontogenetically identified, functionally identified.
And then here is one of the kind of path setting figures in the Friston 2010 paper with a tree
of many different areas to connect to and formalisms to represent as, including probabilistic neural coding,
Bayesian brain, optimal control and value learning.
These three and others can be really seen at play in this 2022 paper.
The kind of inference variational Bayesian methods use rests upon a generative model that expresses a
hypothesis about the generation of sensory input. Perception and behavior can then be read as optimizing
the evidence for a generative model inherent in sensory exchanges for the environment. And that's
the integration of perception and action within a single imperative in terms of if only computation.
And that is described more on this slide, which one can pause and read, but the rest of the quotes are from the paper.

One key formalism that I had no familiarity with before this paper and associated readings and conversations with Ali

was the complete class theorem. So it would be great for anyone who's familiar with complete class theorem to

help a little bit here, but here's some interesting things that I came across.

Crucially from the paper, as a corollary of the complete class theorem, citations 19 through 21,

any neural network minimizing a cost function to be viewed as performing variational Bayesian inference

under some prior beliefs. Here are the citations 19 through 21 from 1947, 1981, 2013.

And also found some interesting resources. So quoting from this less wrong post linked here.

Dutch book defines belief as willingness to bet and money pump defines preference as willingness to pay.

This is in terms of the foundational arguments for a Bayesian epistemological

worldview. Hope that's correct. Again, it'd be good to hear from anyone who knows more here.

But in terms of different ways that one can consider the Bayesian proposition, which perhaps these are even

better to use than referencing any person's last name, because the interesting thing will be the different

lenses that these different framings on Bayesian probabilities are interpreted as, just like what a

p-value is interpreted as in the frequentist world, kind of analogously, like a Bayes factor interpretation.

So Dutch book defines belief as willingness to bet and money pump defines preference as willingness to pay.

In doing so, both arguments put the justification of decision theory into hypothetical exploitation

scenarios, which are not quite the same as the actual decisions we face. If these were the best

justifications for consequentialism we could muster, I would be somewhat dissatisfied, but would likely leave

it alone. Fortunately, a better alternative exists, the complete class theorem.

So here's an image from that post. Possible world states, observation likelihood maps to the possible observations,

hashtag A matrix. Then decision-making rule, maybe probabilistic, action selection as inference,

A possible actions, and here we commonly see the loop being closed from actions to world states like through

the B matrix, changing how the world changes through time. And here, these are both going to be mapped to a loss

function, L, a real valued score from taking an action A when the world turned out to be theta, realized theta.

So that's quite an interesting framing. And there were some other useful posts online.

And from this ZHAT blog, the argument boils down to, if you agree with expected utility as your objective,

then you have to be Bayesian. In a nutshell, a strategy is inadmissible if there exists another strategy that is as good in all

situations and strictly better in at least one. If you want your strategy to be admissible,

it should be equivalent to a Bayes estimator. Complete class theorems. Only Bayes strategies are admissible,

and admissible strategies are Bayes.

So there's probably a lot of interpretations of this, but it seems kind of related to optimal control or Bayes optimal inference, perhaps a little bit more keenly. So these are some next two slides were contributed by Ali.

So if you or if anyone familiar in this area wants to come on in the discussion to be great.

But Ali pointed me to this book, Fundamentals of Bayesian Epistemology by Teitelbaum 2022, table of contents shown here.

And some challenges and objections to Bayesian epistemology are listed, which can be read.

And then also there's some quite interesting logical structures, which can be read here in terms of their premise,

theorem and conclusion in the areas of arguments for Bayesianism being representation theorem, argument for probabilism,

the Dutch book argument for probabilism, and the gradational accuracy argument for probabilism.

So pretty interesting. Back to the paper. They wrote,

We have previously introduced a reverse engineering approach that identifies a class of biologically

plausible cost functions for neural networks. Citation 22. Previous paper of Isomura and Friston.

2020. The paper was reverse engineering neural networks to characterize their cost functions.

The abstract shown here, but this letter considers a class of biologically plausible cost functions

for neural networks. Using generative models based on partially observable Markov decision processes,

POMDP. We show that neural activity and plasticity perform Bayesian inference and learning,

respectively by maximizing model evidence. So this is a precursor paper from 2020 that is cited and

foundational for the 2022 paper. Here's some figures from that paper.

They have comparisons among a Markov decision process scheme and a neural network.

For example, a Markov decision process scheme expressed as a Forney factor graph

based on the formulation in Friston. So here is the Markov decision process as a Forney factor graph. And on the right side is the neural network with the hidden state sensory inputs and neural activity and some figures and concordances.

And they wrote,

In this context, the neural network can in principle be used as a dynamic causal model to estimate threshold constants and implicit priors.

This reverse engineering speaks to estimating the priors used by real neuronal systems under ideal Bayesian assumptions, sometimes referred to as meta Bayesian inference.

So they're laying out their research agenda and much of the work is going to reference this earlier paper and other work.

And then there's also some new integrations and especially the maze simulation in this paper and probably more.

So let's find out.

Referring to the earlier paper,

This foundational work identified a class of cost functions for single layer feed forward neural networks of rate coding models with a sigmoid or logistic activation function.

We subsequently demonstrated that mathematical equivalence between the class of cost functions for such neural networks and variational free energy under a particular form of the generative model, which is similar broadly to what the 22 paper does too.

This equivalence licenses variational Bayesian inference as a fundamental optimization process that underlies both the dynamics and function of such neural networks.

However, it remains to be established whether the active inference is an apt explanation for any given neural network that actively exchanges with its environment.

In this paper, we address this inactive or control aspect to complete the formal equivalence of neural network optimization and the free energy principle.

So this earlier work was not including action.

And so this paper's key addition, as I understand it, hopefully, is that they bring action more formally into the model.

And it'd be interesting to know, like, what else is that change associated with or what else happens or doesn't.

Here's some more text about the approach that they're going to take, and many of these details are going to be addressed later on.

Here's Table 1, Glossary of Expressions.

So these will also be broadly addressed later on.

Maybe it's useful, though, just to read the first one, a canonical neural network.

In this work, a canonical neural network is defined by differential equations of neural activity

derived as a reduction of realistic neuron models through some approximations,

which give a network of rate-coding neurons with a sigmoid activation function.

In particular, we consider networks comprising a middle layer that involves recurrent connections

and the output layer that provides feedback responses to the environment.

So some category of neural network architectures or anatomies, physiologies, whatever,

with sensory, cognitive, and action-like features in an environment or a generative process.

So the results section.

They write,

In Overview of Equivalence Between Neural Networks and Variational Bayes,

First, we summarize the formal correspondence between neural networks and variational bayes.

A biological agent is formulated here as an autonomous system comprising a network of rate-coding neurons,

figure 1a.

So here's a small figure 1a.

We'll see it larger.

Based upon the assumptions, which will be brought up later on in a different fuller form,

the update rule for the ith component of phi,

which is the internal states, so the cognitive states,

and the output layer, y.

So middle layer, x, and output layer, y.

These can be seen as the cognitive and the action-selective aspects,

which are the autonomous states,

in contrast to the particular states of the active inference entity,
which is the whole blanket,
and the cognitive states, so including the sensory states.
But the sensory states can't be directly controlled.
However, action can influence them.
And so that's what closes the causal loop.
And that set of the particular states,
comprising the autonomous states,
the update rule for the ith component of phi
is derived as the gradient descent on the cost function.
So the change in that, phi sub i,
is proportional to a derivative on the loss function.
This determines the dynamics of neural networks,
including their activity and plasticity.
So L, common loss function.
O, observations, sensory states.
Phi, internal states comprising the middle and output layer,
neural activity, that's the rate coding part,
and synaptic strengths, W, the parameterization.
And other free parameters, including that,
that characterize L.
Then there's the output layer activity,
Y, is determining the network's actions or decisions,
delta.
O, X, and Y.
Y is the outputting action, environment,
generative process, hidden state,
passing observations to the sensory, cognitive,
active layers of the neural network.
And that is analogous topologically
to the variational Bayesian formulation
in figure 1b.
And so, left side, gradient descent
on a neural network cost function L
determines the dynamics of neural activity and plasticity.
Thus, L is sufficient to characterize
the neural network.
And then, being shown next to,

on the right side,
variational free energy minimization
allows an agent to self-organize,
to encode the hidden states of the external milieu,
and to make decisions minimizing future risk.
Here, variational free energy F
is sufficient to characterize the inferences
and behaviors of the agent.
So, sentence structure, parallelism,
and these two schemes or approaches
being linked and applied
is a focus of this area.
So, neural networks are minimizing the cost function L.
In contrast, variational Bayesian inference
depicts a process of updating
the prior distribution of external states,
P of theta,
to the corresponding posterior distribution,
Q of theta,
based upon a sequence of observations.
And we see other familiar terms
from discussions on variational Bayes,
like minimization of surprise.
And there's some more model details
shown here.
A few points are about
choosing the family of distributions
that one is doing variational inference with.
And that allows for the gradient update rules
in that family of distributions
to be made simpler.
For example, choosing a linear regression
with an L2 norm,
at the very least,
you can say that it has a simple decision rule
for being fit.
I'm sure there's like also counter arguments
and other algorithms that are better and so on.

But just to say that
a simple decision rule
in a known family of distributions
can often be good enough
as an approximation,
if not more.
Crucially, according to the complete class theorem
from earlier,
a dynamical system
that minimizes its cost function
can be viewed as performing
Bayesian inference
under some generative model
and prior beliefs.
The complete class theorem
just goes on to describe it
ensures the presence
of a generative model
that formally corresponds
to the above-defined neural network
characterized by L.
Hence, this speaks to the equivalence
between the class of neural network
cost functions
and variational free energy
under such a generative model.
Equation 1.
Loss function
on the time series of observations
comma parameters
three lines
is defined as
F
the free energy function
on
the same
O
through time

comma parameters.
So there's a
parallel being
shown
slash defined
wherein the
internal states
of a network
phi
encode
or parameterize
the posterior expectation
theta.
This mathematical equivalence
means that an arbitrary
neural network
in the class
under consideration
is implicitly
performing active inference
through variational
free energy minimization
and more is written
but this is one of many
statements
that will be
made
corresponding
to
correspondences
between
neural network
loss function
generative model
active
free energy
minimization
and associated

formalisms.
Note that being able
to characterize
the neural network
in terms of
maximizing model evidence
lends it
an explainability
in the sense that
internal neural network
states
and parameters
encode
Bayesian beliefs
or expectations
about the causes
of observations.
In other words
the generative model
explains how outcomes
were generated.
However
the complete class
theorem
does not specify
the form
of a generative model
for any given
neural network.
To address this issue
in the remainder
of the paper
we formulate
active inference
using a particular
form of partially
observable
Markov decision

processes
POMDP
models
whose states
take binary values.
So this is one
of several
simplifications
is one way to see it
or areas for
later generalization.
Figure 2
In this section
we define a generative
model and ensuing
variational free energy
that corresponds
to a class of
canonical neural networks
that will be considered
in the subsequent section.
The external milieu
is expressed
as a discrete state space
in the form
of a POMDP
figure 2.
So
to make
figure 2
larger
there's a lot
to probably say
about this
figure.
I'll just
note
that the

caption
is informative
and
some of the
highlighted lines
it'll be awesome
to have the author
and other
people who's familiar
with some
of the differences
and symmetries
between the
for example
bottom
and top
above and below
the observations
and also
about the
role of time
and what
these two risks
are
what is
fictive causality
here's more
details on that
POMDP
then
equation 2
this is in the
section
active inference
formulated using
a postdiction
of past decisions
so if it was a

prediction of
future decisions
it's the
opposite
a postdiction
of past decisions
hence
we define
the generative
model as follows
P
O
delta
S
gamma
theta
which is
the model
of
observations
decisions
observations
decisions
hidden states
risk
and theta
theta
equals
a
b
and c
constitutes
the set of
parameters
and
more details
are provided
this is

the familiar
notation
with
some
slight
differences
that'll be
described
and equation
2
reflects
the
factorizability
and the
dimensions
and the
kind of
computational
tractability
expression
so
that would be
interesting to
also learn
about
like
under what
conditions
can
this
joint
distribution
be
factorized
under what
simplifications
or
constructions

the
agent
is
making
decisions
to
minimize
a
risk
function
capital
gamma
that's
on the
bottom
of
figure
two
fictive
causality
leading to
this
gamma
coming
from
that
gamma
equation
three
is
shown
and
to
characterize
the
optimal
decisions
as

minimizing
expected
risk
in our
POMDP
model
we use
a
fictive
mapping
from
the
current
risk
gamma
to
past
decisions
that's
that
retroactive
fictive
causality
although
this is
not
the
true
causality
in
the
real
generative
process
that
generates
sensory
data

here
we
intend
to
model
the
manner
that
an
agent
subjectively
evaluates
its
previous
decisions
after
experiencing
their
consequences
this
fictive
causality
is
expressed
in
the
form
of
a
categorical
distribution
so
it
could
be
other
families
hypothetically

but
this
equation
3
is
describing
fictive
causality
and
this
interesting
sign
indicates
the
element-wise
division
operator
also
note
throughout
the
manuscript
the
overlying
variable
indicates
1
minus
that
variable
so
gamma
bar
equals
1
minus
gamma
or

it can
be
understood
as
the
complement
of
a
statistical
probability
probability
of
a
happening
and
the
probability
of
not
a
happening
in
that
one
way
importantly
the
agent
needs
to
keep
selecting
good
decisions
while
avoiding
bad
decisions

to
this
end
equation
3
supposes
that
the
agent
learns
from
the
failure
of
decisions
by
assuming
that
the
bad
decisions
were
sampled
from
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opposite
of
the
optimal
policy
mapping
some
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details
and
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convention

active
inference
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c
to
denote
the
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preference
that's
how
we've
seen
the
c
variable
in
many
models
as
preferences
prior
preference
this
work
uses
c
to
denote
a
mapping
to
determine
a
decision
depending
on
the

previous
state
herein
the
prior
preference
is
implicit
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the
risk
function
due
to
construction
c
does
not
explicitly
appear
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the
inference
thus
it
is
omitted
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following
formulations
so
that's
a key
point
about
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notation

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variable
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equation
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variational
bayesian
inference
refers
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optimizes
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belief
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of
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approximation
q
of
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factorization
representation
of
the
q
of
theta
variational
and
another
notation
note
throughout
the
manuscript
bold
case
variables
such
as
bold
s
sub
tau
denote
the
posterior
expectations
of
the
corresponding

italic
case
random
variables
and
some
more
details
about
the
model
for
example
for
simplicity
here
we
suppose
that
state
and
decision
priors
d
and
e
are
fixed
so
one
could
imagine
they
don't
have
to
be
but

for
simplicity
they
will
be
here
under
the
above
defined
generative
model
and
posterior
beliefs
the
ensuing
variational
free
energy
is
analytically
expressed
as
follows
equation
five
this is
a
variational
free
energy
f
and
recall
equation
one
that

it
is
being
connected
to
juxtaposed
with
etc
the
loss
function
the
gradient
descent
on
variational
free
energy
updates
the
posterior
beliefs
about
hidden
states
s
decisions
delta
and
parameters
theta
the
optimal
posterior
beliefs
that
minimize
variational

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implicit
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descent
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the
change
in
hidden
state
through
time
equals
zero
and
some
more
definitions
all of
them

are
zero
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might
be
an
o
but
!
they're
all
zero
and
this
is
saying
the
ball
rolls
to
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bottom
of
the
hill
in
this
gradient
descent
and
when
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rate
of
change
is
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is
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fixed
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attractor
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dynamic
and
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to
incrementally
fit
statistical
models
like
the
loss
function
is
used
to
incrementally
fit
neural
networks
to
explicitly
demonstrate
the
formal

correspondence
with
the
cost
function
for
neural
networks
considered
in the
next
section
we
further
transform
the
variational
free
energy
as
follows
some
details
using
these
relationships
equation
five is
transformed
into the
form shown
in figure
three
see the
method
section
variational
free

energy
for
further
details
in
what
follows
we
demonstrate
that
the
internal
states
of
canonical
neural
networks
encode
posterior
beliefs
here's
figure
three
on the
top
from
the
caption
figure
three
is
the
mathematical
equivalence
between
variational
free
energy

and
neural
network
cost
functions
depicted
by
one-to-one
correspondence
of their
components
top
variational
free
energy
transformed
from
equation
five
using
the
Bayes
theorem
and
foreshadowing
equation
seven
using
this
relationship
equation
seven
is
transformed
into
the
form
presented

at
the
bottom
of
the
figure
so
here's
f
variational
free
energy
and
I
the
neural
network
cost
function
and
different
elements
as
they're
represented
here
with
some
resonances
and
concordance
and
beyond
which
we can
explore
they're
being

shown
to be
equivalent
and
it was
in the
conversation
preparing
for this
dot
zero
with
Dean
where
we
saw
this
as
some
chromosomes
in
this
section
canonical
neural
networks
perform
active
inference
in
this
section
we
identify
the
neuronal
substrates
that

correspond
to
components
of
the
active
inference
scheme
!
We
consider
a
class
of
two
layer
neural
networks
with
recurrent
connections
in the
middle
layer
figure
1a
those
are
those
connections
with
loops
recurrent
in the
middle
layer
the
modeling

of the
networks
in this
section
referred
to as
canonical
neural
networks
is
based
on the
following
three
assumptions
that
reflect
physiological
knowledge
1
gradient
descent
on a
cost
function
L
determines
the
updates
of
neural
activity
and
synaptic
weights
method
section
neural

networks
neural
two
assumption
two
neural
activity
is
updated
by the
weighted
sum
of
inputs
and
its
fixed
point
is
expressed
in a
form
of
the
sigmoid
or logistic
function
and
assumption
three
a
modulatory
factor
mediates
synaptic
plasticity
in a
post hoc

manner
they
based

upon
assumption
two
which
is
neural
activity
is
updated
by
the
weighted
sum
of
inputs

and
its
fixed
point
is
expressed
in the
form
of a
sigmoid
or logistic
function
based
on
assumption
two
we
formulate

neural
activity
in the
middle
layer
and
output
layer
the
autonomous
states
as
follows
equation
six
without
loss
of
generality
equation
six
can be
cast
as
the
gradient
descent
on
cost
function
I
such
a
cost
function
can be
identified
by

simply
integrating
the
right-hand
side
of
equation
six
with
respect
to
x
and
y
consistent
with
previous
treatments
citations
because
neural
activity
and
synaptic
plasticity
minimize
the
same
cost
function
I
the
derivatives
of
I
must
generate
the

modulated
synaptic
plasticity
under
these
constraints
reflecting
assumptions
one
through
three
a
class
of
cost
functions
is
identified
as
follows
equation
seven
loss
function
awesome
to hear
somebody
read
this
directly
so
there's
a
firing
rate
aspect
that's
related

to
neural
firing
in the
short
term
more
perceptual
aspect
of
function
and
there's
a
slower
neurotransmitter
neural
glial
factor
mediated
learning
over
perhaps a
different
timescale
and with
some
different
features
and
they're
being
included
as a
joint
model
of
inference

and
here
that
is
being
connected
to
doing
inference
on
action
synaptic
plasticity
in neural
networks
so
synaptic
plasticity
rules
conjugate
to the
above
rate
coding
model
can
now
be
expressed
as a
gradient
descent
on
the
same
cost
function
L

according
to
assumption
one
equations
eight
and
nine
showing
that
neural
networks
can
integrate
those
modes
of
firing
rate
like
and
synaptomodulatory
like
neural
networks
cost
functions
and
variational
free
energy
based
on
the
above
considerations
we
now

establish
the
formal
correspondence
between
the
neural
network
cost
function
and
variational
free
energy
using
under
the
aforementioned
three
minimal
assumptions
we
identified
the
neural
network
cost
function
as
equation
seven
equation
seven
can
be
transformed
into
the

form
shown
in
figure
three
bottom
using
sigmoid
functions
of
synaptic
strengths
so
equation
seven
loss
function
transformed
into
the
form
shown
in
figure
three
bottom
hence
this
class
of
cost
functions
for
canonical
neural
networks
is
formally

homologous
to
variational
free
energy
under
the
particular
form
of
the
POMDP
generative
model
defined
in
the
previous
section
we
obtain
this
result
based
on
analytic
derivations
without
reference
to the
complete
class
theorem
thereby
confirming
the
proposition
of

equation
one
loss
function
and
free
energy
this
in
turn
suggests
that
any
canonical
neural
network
in
this
class
is
implicitly
performing
active
inference
table
two
summarizes
the
correspondence
between
the
quantities
of
the
neural
network
and
their

homologues
in
variational
base
so
two
a
great
concordance

table
on the
left
side
neural
network
formation
on the
right
side
variational
base
formulation
so
just
like
in
figure
three
we
had
the
two
long
lines
then
table
two

we
have
rotated
90
degrees
and
the
variables
for
different
parts
of
these
models
are
laid
next
to
each
other
so
what
papers
of
neural
networks
can
we
make
active
models
for
is it
already
done
or
is there
just

one
script
that
needs
to be
done
and
then
conversely
what
interesting
variational
Bayesian
models
have
interesting
applications
or
some
other
value
or
information
gain
from
being
cast
as
neural
networks
or
integrating
neural
networks
into
what
had
previously

been
only
analytical
variational
Bayesian
models
as well
as the
empirical
data
fitting
aspect
which
is
related
that's
highlighted
by the
authors
so
in
summary
when
a
neural
network
minimizes
the
cost
function
with
respect
to
its
activity
and
plasticity
the

network
self
organizes
to
furnish

responses
that
minimize
a
risk
implicit
in
the
cost
function
this
biological
optimization
is
identical
to
variational
free
energy
minimization
under
a
particular
form
of
the
POMDP
model
hence
this
equivalence
indicates

that
minimizing
the
expected
risk
through
variational
free
energy
minimization
is an
inherent
property
of
canonical
neural
networks
featuring
delayed
modulation
of
Hebbian
plasticity
!
Okay
brief
seconds
to
view
some
image
memes
and
take
a
breath
and
a

stretch
on
the
left
side
the
top
panel
says
every
neural
network
is
implicitly
performing
active
inference
question
mark
that's
awesome
in
the
middle
image
meme
one
simply
makes
a
neural
network
and
it
is
implicitly
performing
active

inference
also
question
mark
and
then
in
the
right
image
meme
there
are
two
pieces
of
paper
neural
network
and
implicit
performance
of
active
inference
corporate
needs
you to
find
the
differences
between
this
picture
and
this
picture
and

here
is
the
paper
they're
the
same
picture
so
the
previous
sections
have
successfully
extended
the
variational
base
meets
inference
neural
network
result
from the
2020
paper
with an
action
loop
and a
loss
function
only on
the
autonomous
states
so
that was

the
conceptual
and
technical
feature
that
this
paper
brought
in
again
it'll
be
awesome
to
hear
the
author
describe
it
more
and
differently
and
then
in
this
part
of
the
paper
they
turn
from
that
kind
of
analytical

theoretical
towards
some
numerical
simulations
in
MATLAB
here
we
demonstrate
the
performance
of
canonical
neural
networks
using
maze
tasks
as
an
example
of
a
delayed
reward
task
the
agent
comprised
the
aforementioned
canonical
neural
networks
figure
4a
thus

it
implicitly
performs
active
inference
by
minimizing

variational
free
energy
so
now
some
entity
or
agent
is
going
to
be
constructed
and
numerically
simulated
to
support
some
of
the
aspects
of
their
model
and
point
towards
utility

in
other
settings
so
here's
figure
4
simulation
of
neural
networks
solving
maze
tasks
!
of
the
agent
sensory
input
layer
O
of
T
synaptic
weights
into
the
middle
layer
X
internal
states
and
the
output
layer
Y

decision
action
with
risk
gamma
of
T
the
sensation
comes in
a neural
network
then
outputs
a
decision
there's
some
task
that
the
entity
must
perform
which
is
to
move
towards
the
goal
from
the
start
across
this
lateral
maze

so
one
can
imagine
that
if
it
were
just
a
hallway
with
no
maze
features
simple
strategies
would be
very
overfit
but
effective
like
always
move
simply
towards
the
goal
whereas
in
more
complex
settings
which
this
is an
example

of
where
there's
uncertainty
as well
as many
local
optima
so
one
has
to
sometimes
take
one
or
two
or
three
or
four
or
more
steps
or
however
many
to
get
closer
to
the
goal
and
sometimes
may
not
know

for
example
how
long
those
backtracking
situations
are going
to be
and all
these other
complexities
for which
this maze
example
is
symbolic
of
so
maybe
there's
some
interesting
mythos
maze
connections
and
computational
and
here's
some
performance
measures
on
the maze
task
with
the

neural
network
entity
this way
before
training
the agent
moved to
a random
direction
in each
step
resulting
in a
failure
to
reach
the
goal
position
within
the
time
limit
during
training
the
neural
network
updated
synaptic
strengths
depending
on its
neural
activity
and
ensuing

outcomes
i.e.
risk
this
training
comprised
a cycle
of
action
and
learning
phases
this
treatment
renders
neural
activity
and
adaptive
behaviors
of the
agent
highly
explainable
and
manipulatable
in terms
of the
appropriate
prior
beliefs
implicit
and firing
thresholds
for a
given
task
or

environment
in other
words
these
results
suggest
that
firing
thresholds
are the
neuronal
substrates
that encode
state and
decision
priors
as predicted
mathematically
big if
true
further

when the
updating
of
parameters
is slow
across
these
two
now
linked
domains
parameters
of the
variational
bayesian
autonomous

state
inference
and

the
neural
network
loss
function
parameters
when
the
updating
of
these
parameters
are
slow
in
relation
to
the
experimental
observations
the
parameters
can be
estimated
through
bayesian
inference
based
on

empirically
observed
neuronal
responses
see
method
section
data
analysis
for
details
using
this
approach
we
estimated
implicit
prior
e
which is
encoded
by
psi
from
sequences
of
neural
activity
generated
from
the
synthetic
neural
networks
used
in
the
simulations

reported
in
figure
four
we
confirmed
that
this
estimator
was a
good
approximation
to
the
true
e
so
that's
also
pretty
interesting
this
is
showing
they
don't
just
lay
out
this
architecture
and
show
that
it's
possible
to
fulfill

this
maze
task
with
the
best
whatever
it's
not a
classification
accuracy
imperative
alone
they're
describing
that
from
empirically
observed
neuronal
responses
which is
to say
the
experimenter's
observation
the
sequences
of neural
activity
generated
from the
synthetic
neural
networks
used
in
that

figure
numerically
so
statistically
that
estimator
was a
good
approximation
to the
true
e
figure
5
a
so
here's
figure
5
estimation
of
implicit
priors
enables
the
prediction
of
subsequent
learning
so
that's
pretty
interesting
and
will be
great
to hear
what

each
of the
axes
are
and
what
all the
variables
mean
with
this
set up
in
place
they
did
some
numerical
validation
and
talked a
little bit
more
about
fitting
data
from
this
simulation
model
here's
the
discussion
section
so
the first
paragraph
of the

discussion
biological
organisms
formulate
plans
to
minimize
future
risks
in
this
work
we
captured
this
characteristic
in
biologically
plausible
terms
under
minimal
assumptions
we
derive
simple
differential
equations
that
can
be
plausibly
interpreted
in
terms
of
a
neural

network
architecture
that
entails
degrees
of
freedom
with
respect
to
certain
free
parameters
e.g.
firing
threshold
these
free
parameters
play
the
role
of
prior
beliefs
in
variational
Bayesian
formulation
thus
the
accuracies
of
inferences
and
decisions
depend
upon

prior
beliefs
implicit
in
neural
networks
and
here's
some
more
stable
diffusion
ant
neural
network
ant
brain
neural
network
so
some
more
summarization
of what
they did
based
on the
view
of the
brain
as an
agent
that
performs
Bayesian
inference
neuronal
implementations

of Bayesian
belief
updating
have been
proposed
which
enables
neural
networks
to
store
and
recall
spiking
sequences
8
learn
temporal
dynamics
and
causal
hierarchy
9
extract
hidden
causes
10
solve
maze
tasks
11
and
make
plans
to
control
robots
12

so
citations
8-12
listed
here
in
these
approaches
the
update
rules
are
generally
derived
from
Bayesian
cost
functions
e.g.
variational
free
energy
however
the precise
relationship
between
these
update
rules
and the
neural
activity
and
plasticity
of
canonical
neural
networks

has yet
to be
fully
established
we
identified
a one-to-one
correspondence
between neural
network
architecture
and a
specific
POMDP
implicit
in that
network
equation
2
speaks
to a
unique
POMDP
model
consistent
with the
neural
network
architecture
defined
in
equation
6
where
their
correspondences
are summarized
in table

2
and
the
figures
this
means
that
our
scheme
can be
used
to
identify
the
form
of
the
POMDP
given
an
observable
circuit
structure
moreover
the
free
parameters
that
parameterize
equation
6
can be
estimated
using
equation
24
this
means

that
the
generative
model
and
ensuing
variational
free
energy
can
in
principle
be
reconstructed
from
empirical
data
this
offers
a
formal
characterization
of
implicit
Bayesian
models
entailed
by neural
circuits
thereby
enabling
a
prediction
of
subsequent
learning
so
what

can
be
done
with
this
what
is
this
new
what
is
new
here
does
this
second
selection
fully
establish
the
precise
relationship
between
these
update
rules
and
the
neural
activity
and
plasticity
of
canonical
neural
networks
here
is a

discussion
section
on
Hebbian
plasticity
neurotransmitters
and glia
with a lot
of
citations
listed
and
here
is just
one
interesting
follow-on
discussion
from
the
computational
aspects
neuro
cognitive
and
computational
aspects
of
Hebbian
plasticity
is
these
modulations
have been
observed
empirically
with
various

neuromodulators
and
neurotransmitters
such as
dopamine
noradrenaline
muscarine
and
GABA
as well
as
glial
factors
so
here's
a
picture
by
Alexandra
Mikhailova
cultured
astrostites
release
signaling
and
growth
factors
that
regulate
proper
neuronal
development
so
cool
glial
pick
dopamine
and

reinforcement
learning
in particular
a delayed
modulation
of
synaptic
plasticity
is well
known
with
dopamine
neurons
citations
35
through
37
those
citations
are
listed
here
this
speaks
to a
learning
scheme
that
is
conceptually
distinct
from
standard
reinforcement
learning
algorithms
such
as

the
temporal
difference
learning
with
actor
critic
models
based
on
state
action
value
functions
please
see
the
previous
work
citation
50
for a
detailed
comparison
between
active
inference
and
reinforcement
learning
that
is
Sajid
Ball
Parr
and
Friston
active

inference
demystified
and
compared
from
2021
and
that's
also
active
model
stream
number
2.1
we
mathematically
demonstrated
that such
plasticity
enhances
the
association
between
the
pre
post
mapping
and
the
future
value
of
the
modulatory
factor
where
the
latter

is
cast
as
a
risk
function
this
means
that
post
synaptic
neurons
self
organize
to
react
in
a
manner
that
minimizes
future
risk
so
the
neural
network
had
three
layers
it's
the
second
and
the
third
layer
not

the
initial
sensory
layer
but
the
second
and
the
third
layer
the
cognitive
and the
active
states
which are
the
autonomous
states
of

the
particular
states
so
that's
quite
interesting
the
self
organization
of
post
synaptic
neurons
to
react

in a
manner
that
minimizes
future
risk
crucially

this
computation
corresponds
formally to
variational
Bayesian
inference
under a
particular
form of
POMDP
generative
models
suggesting
that the
delayed
modulation
of
Hebbian
plasticity
is a
realization
of
active
inference
and
regionally
specific
projections
of

neuromodulators
may allow
each
brain
region
to
optimize
activity
to
minimize
risk
and
leverage
a
hierarchical
generative
model
implicit
in
cortical
and
subcortical
hierarchies
this is
reminiscent
of
theories
of
neuromodulation
and
meta-learning
developed
previously
citation
52
doya
2002
meta-learning

and
neuromodulation
cool
our work
may be
potentially
useful
when
casting
these
theories
in terms
of
generative
models
and
variational
free
energy
minimization
they
then
return
the
discussion
to
the
complete
class
theorem
and
neural
networks
so
the
complete
class
theorem

same
citations
from
before
ensures
that
any
neural
network
whose
activity
and
plasticity
minimize
the same
cost
function
can be
cast
as
performing
Bayesian
inference
however
identifying
the implicit
generative
model
that
underwrites
any
canonical
neural
network
is a
more
delicate
problem

because
the
theorem
does not
specify
a form
of the
generative
model
for a
given
canonical
neural
network
which is
pretty
interesting
is that
to say
that the
form of
the
generative
model
as
modeled
is
different
in what
ways
from
the
given
canonical
neural
network
the
posterior

beliefs
are
largely
shaped
by
prior
beliefs
making
it
challenging
to
identify
the
generative
model
by
simply
observing
systemic
dynamics
to
this
end
it is
necessary
to
commit
to
a
particular
form
of
the
generative
model
and
elucidate
how

posterior
beliefs
are
encoded
or
parameterized
by the
neural
network
states
this
work
addresses
these
issues
by
establishing
a
reverse
engineering
approach
to
identify
a
generative
model
implicit
in a
canonical
neural
network
thereby
establishing
one-to-one
correspondences
between
their
components

remarkably
a network
of rate
coding
models
with
sigmoid
activation
function
formally
corresponds
to a
class
of
POMDP
models
which
provide
an
analytically
tractable
example
of the
present
equivalence
please
refer to
the
previous
paper
citation
22
for
further
discussion
so
some
of

the
analytical
details
especially
on the
inferential
side
cognitive
side
were
captured
in the
earlier
paper
this
paper
goes
further
into
mapping
the
potentially
necessary
and
sufficient
aspects
of
the
particular
entity
which is
to say
the risk
minimizing

features
of
the

autonomous
states
with respect
to the
entire
particular
states
including
sensory
states
connecting
that
back of
the
envelope
verbally
expressible
formulation
to
some
complete
class
of
neural
networks
so
what's
outside
of the
complete
class
and why
and then
what
groups
within
the
class

have
special
or
different
features
it is
remarkable
that the
proposed
equivalence
can be
leveraged
to identify
a generative
model
that an
arbitrary
neural
network
implicitly
employs
this
contrasts
with
naive
neural
network
models
that
address
only
the
dynamics
of
neural
activity
and
plasticity

if the
generative
model
differs
from the
true
generative
process
that
generates
the
sensory
input
inferences
and decisions
are biased
i.e.
suboptimal
relative to
Bayes'
optimal
inferences
and decisions
based upon
the right
sort of
beliefs
prior
beliefs
in
general
the
implicit
priors
may
or
may
not

be
equal
to
the
true
priors
thus
a
generic
neural
network
is
typically
suboptimal
nevertheless
these
implicit
priors
can be
optimized
by
updating
free
parameters
e.g.
threshold
factors
by
psi
based
on the
gradient
descent
on
cost
function
L
by

updating
the
free
parameters
the
network
will
eventually
in
principle
become
Bayes
optimal
for
any
given
task
in
essence
when
the
cost
function
is
minimized
with
respect
to
neural
activity
synaptic
strengths
and
any
other
constants
that
characterize

the
cost
function
the
cost
function
becomes
equivalent
to
variational
free
energy
with
the
optimal
prior
beliefs
so
the
cost
function
for
the
neural
network
activity
and
synaptic
strengths
underlying
the
loss
function
are
equivalent
to
the
kind

of
gradient
descent
enabled
variational
free
energy
minimization
under
the
Bayes
optimality
scenario
from a
risk
perspective
simultaneously
the
expected
risk
is
minimized
because
variational
free
energy
is
minimized
only
when
the
precision
of
the
risk
gamma
is
maximized

see
method
section
generative
model
for
further
details
very
interesting
they
then
say
the
class
of
neural
networks
we
consider
can
be
viewed
as
a
class
of
reservoir
networks
citation
54
citation
55
here
the
proposed
equivalence
could

render
such
reservoir
networks
explainable
and
may
provide
the
optimal
plasticity
rules
for
these
networks
to
minimize
future
risk
by
using
the
formal
analogy
to
variational
free
energy
minimization
under
the
particular
form
of
POMDP
models
a
clear

interpretation
of
reservoir
networks
remains
an
important
open
issue
in
computational
neuroscience
and
machine
learning
so
from
wikipedia
reservoir
computing
is a
framework
for
computation
derived
from
recurrent
neural
network
theory
that
maps
input
signals
into
higher
dimensional
computational

spaces
through
the
dynamics
of
a
fixed
nonlinear
system
called
a
reservoir
after
the
input
signal
is
fed
into
the
reservoir
which
is
treated
as
a
black
box
a
simple
readout
mechanism
is
trained
to
read
the
state

of
the
reservoir
and
map
it
to
the
output
then
there
are
two
key
benefits
of
this
approach
the
first
key
benefit
of
this
framework
is
that
training
is
performed
only
at
the
readout
stage
as
the
reservoir

dynamics
are
fixed
the
second
is
that
the
computational
power
of
naturally
available
systems
both
classical
and
quantum
mechanical
can
be
used
to
reduce
the
effective
computational
cost
here's
some
stable
diffusion
reservoir
computing
active
inference
neural
network

so
this
would be
interesting
to discuss
and I
remember
some
act
institute
participants
who are
specifically
interested
empirical
analyses
and
hypotheses
they
write
the
equivalence
between
neural
network
dynamics
and
gradient
flows
on
variational
free
energy
is
empirically

testable
using

electrophysiological
recordings
or
functional
imaging
of
brain
activity
so
then
another
summarization
crucially
the
proposed
equivalence
guarantees
that
an
arbitrary
neural
network
that
minimizes
its
cost
function
possibly
implemented
in
biological
organisms
or
neuromorphic
hardware
can be
cast as
performing

variational
Bayesian
inference
so
to
state
it
a few
more
times
in
the
final
paragraph
in
summary
a
class
of
biologically
plausible
cost
functions
for
canonical
neural
networks
can be
cast
as
variational
free
energy
formal
correspondences
exist
between
priors

posteriors
and
cost
functions
this
means
that
canonical
neural
networks
that
optimize
their
cost
functions
implicitly
perform
active
inference
this
approach
enables
identification
of the
implicit
generative
model
and
reconstruction
of
variational
free
energy
that
neural
networks
employ
this

means
that
in
principle
neural
activity
behavior
and
learning
through
plasticity
can be
predicted
under
Bayes
optimality
assumptions
there's
a code
availability
statement
the
MATLAB
scripts
are
available
at
github
in
the
repo
of
the
first
author
and
it
would

be
awesome
for
the
author
or
anyone
to
bring
this
working
MATLAB
script
up
and see
if we
could do
some
real
time
active
inference
methods
then
as
mentioned
earlier
from
the
roadmap
the
methods
are
following
the
discussion
I'll
just

show
the
equations
but not
go into
any
details
because
there's
not
time
nor
familiarity
so
those
who
have
more
of
one
or
the
other
would
be
welcome
to
fill in
some
dots
because
this
is a
really
awesome
and
important
paper

so I
hope
that
it
can
be
interpreted
and
critiqued
and elaborated
on
and so
on
by those
with
familiarity
in
both
sides
of that
free
energy
loss
function
equation
one
generative
model
section
many
details
are
provided
equation
10
is
shown
it's

a
larger
unpacking
of the
generative
model
section
variational
free
energy
many
details
are
provided
equation
11
12
13
14
and
15
section
inference
and
learning
details
are
provided
equations
16
17
18
then
section
on
neural
networks
so

there's
just
some
interesting
writing
here
so
I
wanted
to
highlight
that
in
this
section
we
elaborate
the
one-to-one
correspondences
between
components
of the
canonical
neural
networks
and
variational
bays
via
an
analytically
tractable
model
so
that's
the
figure

three
that
we've
been
returning
to
neurons
respond
quickly
to a
continuous
stimulus
stream
with a
timescale
faster than
typical
changes
in
sensory
input
for
instance
a
peak
of
spiking
neurons
in the
visual
cortex
of
v1
appears
within
50
and
80

milliseconds
after
a
visual
stimulation
citation
6263
which is
substantially
faster
than the
temporal
autocorrelation
timescale
of
natural
image
sequences
about
500
milliseconds
citation
64
65
so
that's
pretty
interesting
what is
the
temporal
autocorrelation
timescale
of
natural
sequences
what
timescale

do
neural
firing
and
neuroplasticity
etc
processes
actually
occur
at
and
when
might
some
type
of
function
at a
given
timescale
be
understood
to be
functional
or not
thus
in other
words
downstream
of the
fact that
neurons
respond
quickly
at a
timescale
faster than
typical

changes in
sensory
input
we consider
the case
where the
neural
activity
converges
to a
fixed
point
given
a
sensory
stimulus
we
note
that the
present
equivalence
is derived
from the
differential
equations
equation
six
but not
from its
fixed
point
thus
the
equivalence
holds
true
irrespective
of the

time
constant
of
neurons
to
rephrase
neural
networks
with a
large
time
constant
!
formally
correspond
to
Bayesian
belief
updating
with a
large
time
constant
which
implies
an
insufficient
coverage
of the
posterior
beliefs
pretty
interesting
related to
learning
rates
and
Bayesian

updating
rates
and all
of the
nooks and
crannies
of Bayesian
inference
and the
challenges
associated
with dynamic
uncertain
rugged
fitness
and
free energy
landscapes
these
optimization
challenges
which are
addressed
differently
methodologically
culturally
etc.
in the
variational
Bayesian
and in
the
neural
cases
they
have to
deal with
time

and so
all of
those
different
nooks and
crannies
mentioned
like
catastrophic
learning
catastrophic
forgetting
simply
memory
anticipation
attention
local
maxima
choosing
when to
play
all these
higher order
questions
are
connected
here
does that
make it
what kind
of a
problem
space
now
or just
what kind
of space
pretty

interesting
and
equation
19
20
21
22
23
they
have
some
more
details
on
the
simulation
maybe
we
could
see
the
simulation
go
and
in
the
last
section
data
analysis
so
this
is
kind
of
returning
to
that

point
about
the
time
constants
in
Bayesian
and
neural
network
systems
when the
belief
updating
of implicit
priors
d and e
is slow
in relation
to
experimental
observations
d and e
which are
encoded
by
phi and
psi
can be
viewed as
being
fixed
over a
short
period
of
time
as

an
analogy
to
homeostatic
plasticity
over
longer
timescales
66
homeostatic
plasticity
in the
developing
nervous
system
2004
very
interesting
in light
of our
recent
discussions
on
allostasis
and other
topics
thus
phi and
psi
can be
statistically
estimated
by a
conventional
Bayesian
inference
or
maximum

likelihood
estimation
given a
flat
prior
based
on
empirically
observed
neuronal
responses
in this
case
the
estimators
of
phi and
psi
are
obtained
as
follows
nice
equation
number
24
mentioned
earlier
so that
will be
definitely
one to
look into
more
well
I hope
you found
this a

useful
and
interesting
dot
zero
I'm
looking
forward
to the
discussion
with
the
author
and
again
any
other
institute
participants
or
those
with
knowledge
or
strong
feelings
to
express
about
neural
networks
active
inference
applied
active
inference
computational
modeling

of perception
cognition
and action
neural
networks
in the
wild
AI
ethics
all

these
different
areas
can
hopefully
have
a nice
discussion
in
51.1
and
.2
that'll
be the
last
paper
streams
for
2022
there will
still be a
few
guest
streams
upcoming
so
those

discussions
will
close the
paper
focused
prepared
live
streams
for
2022
and
yeah
if you
want to
be more
involved
with live
streams
whenever
you're
listening
to
this
join
or
recommend
someone
to join
or help
in any
number
of other
ways
just
listening
and learning
is
awesome

and we
also
really
look
forward
to
those
who
want
to
help
make
some
of
these
connections
that
are
latent
and
sometimes
exposed
in
these
papers
and
conversations
and
with
a few
motivated
people
who
want
to
connect
for
example

to
the
neural
network
communities
and
those
who
can
facilitate
discussions
on
these
topics
that
would
be
awesome
just
using
my
final
thoughts
on
this
dot
zero
because
it's
always
great
to
have
others
also
join
in
the

preparation
for
the
dot
zero
so
just
want
to
add
that
note
on
this
rare
solo
stream
so
thanks
again
looking
forward
to
talking
or
seeing
you
later
bye
!